

Analog Electronics

1. Amplifier – Increases signal strength.
2. Bandwidth – Range of frequencies amplifier works.
3. Biasing – Setting operating point of device.
4. BJT (Bipolar Junction Transistor) – Current-controlled device.
5. Bypass Capacitor – Stabilizes voltage by bypassing AC.
6. Clamping Circuit – Shifts signal level.
7. Clipping Circuit – Cuts off parts of signal.
8. Common Base – BJT configuration with low input impedance.
9. Common Collector – BJT configuration used as buffer.
10. Common Emitter – BJT configuration with high gain.
11. Coupling Capacitor – Passes AC, blocks DC.
12. Diode – Device allowing current in one direction.
13. FET (Field Effect Transistor) – Voltage-controlled device.
14. Feedback Amplifier – Uses feedback to improve performance.
15. Filter – Smoothens rectified output.
16. Forward Bias – Reduces barrier, allows current flow.
17. Full-Wave Rectifier – Uses both halves of AC.
18. Gain – Ratio of output to input signal.
19. Half-Wave Rectifier – Uses one half cycle of AC.
20. Input Impedance – Opposition to input signal.
21. MOSFET – Type of FET with insulated gate.
22. Oscillator – Generates continuous waveform.
23. Output Impedance – Opposition at output terminal.
24. PN Junction – Boundary between P-type and N-type materials.
25. Q-Point – Stable operating point of transistor.
26. Rectifier – Converts AC to DC.
27. Reverse Bias – Increases barrier, blocks current.
28. Semiconductor – Material with conductivity between conductor and insulator.
29. Voltage Regulator – Maintains constant output voltage.
30. Zener Diode – Maintains constant voltage in breakdown region.

Digital Electronics

1. Adder – Circuit that performs binary addition.
2. Analog Signal – Continuous signal in time and amplitude.
3. AND Gate – Outputs 1 only when all inputs are 1.
4. ASCII Code – Standard binary encoding for characters.
5. BCD (Binary Coded Decimal) – Decimal digits represented in binary form.
6. Binary Number System – Uses only 0 and 1.
7. Boolean Algebra – Mathematical system for logic operations.
8. Buffer – Circuit that strengthens signal without changing it.
9. Clock Signal – Timing signal for sequential circuits.
10. Combinational Circuit – Output depends only on present inputs.
11. Comparator – Compares two binary numbers.
12. Counter – Sequential circuit that counts pulses.
13. Decoder – Converts binary input into one of many outputs.
14. Demultiplexer – Routes one input to multiple outputs.
15. Digital Signal – Discrete signal with finite levels.
16. Encoder – Converts multiple inputs into binary output.
17. Flip-Flop – Basic memory element storing 1 bit.
18. Frequency – Number of cycles per second.
19. Full Adder – Adds three binary bits.
20. Half Adder – Adds two binary bits.
21. JK Flip-Flop – Universal flip-flop with toggle feature.
22. Latch – Level-triggered memory device.
23. Logic Gate – Basic building block of digital circuits.
24. Multiplexer (MUX) – Selects one input from many.
25. NAND Gate – NOT of AND gate output.
26. NOR Gate – NOT of OR gate output.
27. NOT Gate – Inverts input signal.
28. OR Gate – Outputs 1 if any input is 1.
29. Register – Group of flip-flops storing data.
30. Shift Register – Moves data sequentially.